Thirteen Variables from a Weak *abc* Inequality

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Abstract

Assume the weak rational *abc* inequality with exponent 121/100, together with the specified elliptic group, pole-realization and simultaneous prime-pattern inputs. We construct a family of integer polynomials in thirteen total variables. Every sufficiently large fixed odd input shift gives rational input projection exactly $\mathbb{Z}$. Each member has total degree less than 2^{44}. A degree-four rational map gives a logarithmic bound for an odd contact radical from the weak inequality. An eighth-power height comparison then bounds the rational input and one elliptic index together. Only three elliptic points and two pole certificates occur, with six local tests of order one. The coefficient threshold is ineffective. Both the rational *abc* conjecture and the ordinary Szpiro conjecture imply the weak inequality used here.

1 Statement and hypotheses

All variable counts include the input. For a rational number a/b in lowest terms with $b > 0$, write $h(a/b) = \log \max(|a|, b)$. For a nonzero integer, its radical is the product of its distinct prime divisors, with positive sign. Put $\kappa = 121/100$. The additional height hypothesis used in this paper is the following single inequality: there is a constant $K_\kappa \geq 1$ such that

$$c \leq K_\kappa \operatorname{rad}(abc)^\kappa \quad (a + b = c,\ a, b, c \text{ positive and pairwise coprime}). \tag{1}$$

The same inequality, with the largest absolute value on the left, applies to a coprime nonzero signed triple of sum zero, by moving its negative terms to the other side.

Set

$$K = \mathbb{Q}(\sqrt{2}), \quad D = (3 - \sqrt{2})^2, \quad E : y^2 = x^3 - Dx,$$
$$P = \left(\frac{57}{8} - \frac{7}{2}\sqrt{2}, -\frac{165}{8} + \frac{417}{32}\sqrt{2}\right).$$

In addition to (1), retain the following arithmetic inputs with exactly their stated scope in [3, 2].

1. $2E(K) = \mathbb{Z}P$, with the indicated non-torsion generator and the source index-height bound. The direct point chart below therefore describes the nonzero multiples of P.

2. The pole system, for $b \neq 0$ and $B_0 = 17b$,

$$p^2 = B_0(1 - e)(1 + 3e), \qquad B_0 q^2 = (s^2 - B_0 e(B_0 - b^2))(s^2 + 3B_0 e(B_0 - b^2)),$$

excludes negative odd valuations of b outside $\{2, 3, 17\}$. For a prepared input

$$\begin{gathered} 0 < b < 17, \quad b \text{ has square reduced denominator}, \quad b \in \mathbb{Q}_2^{\times 2} \cap \mathbb{Q}_3^{\times 2} \cap \mathbb{Q}_{17}^{\times 2}, \\ v_2(b) \geq 3, \quad v_3(b) \geq 1, \quad v_{17}(b) \geq 2, \end{gathered} \tag{2}$$

the retained restricted-converse and descent hypotheses supply a witness with $pqs \neq 0$. The nonzero-witness consequence and its single conic chart are proved in [2, Lemma 2].

3. The simultaneous prime-pattern input supplies the specified prime contact numerators, Gaussian congruences, conjugate reduction orders, fixed divisibility and avoidance, and compatible real approximation for a fixed finite list of rational contact coefficients. Its precise quantified version and the rational-contact and fixed-avoidance consequences of Sun's theorem are given in [3]. The even number of contacts below permits arbitrary positive rational scales: Section 6 checks the square factors directly. The order of choices is respected there.

The exponent-nineteen contact-height input of the earlier source is not needed here. It is replaced by Lemma 2.1, which uses (1).

The proved one-point case of the aggregate trace-height theorem [5] gives constants $A > 0$, $J \geq 0$, depending only on E, P, such that

$$n^2 \leq Ah(T(n)) + J, \qquad T(n) = \mathrm{Tr}_{K/\mathbb{Q}}\, x([n]P) \quad (n \in \mathbb{Z} \setminus \{0\}). \tag{3}$$

This applies von Buhren's theorem after checking the absence of constant abelian translates for the fixed trace function. Its additive constant is ineffective. It is a proved height theorem, not an additional conjecture. For a nonzero multiple of P, the abscissa is a positive square in both real embeddings; the curve equation then gives $x > 3 \mp \sqrt{2}$ in the respective embeddings, hence $T(n) > 6$.

Theorem 1.1. *Under the assumptions above, there is a computable family*

$$F_B \in \mathbb{Z}[X, Y_1, \ldots, Y_{12}], \qquad B \geq 3 \text{ odd},$$

with the following properties.

1. *Every member is given by a complete finite integer coefficient specification and satisfies*

$$\deg F_B \leq 12339485119488 < 2^{44}.$$

2. *There exists a constant B_* such that every odd $B \geq B_*$ has rational input projection exactly $\mathbb{Z}$.*

3. *No numerical sufficient value of B_* is determined. The supplied member $B = 3$ is a coefficient-verification fixture, not a polynomial certified to define $\mathbb{Z}$.*

The integer B is a fixed coefficient parameter and is never a quantified coordinate.

Corollary 1.2. *Retain the three source arithmetic inputs above. Either of the following conjectures implies the conclusion of Theorem 1.1:*

1. *the ordinary rational abc conjecture;*

2. *the ordinary Szpiro conjecture: for every $\epsilon > 0$ there is a constant C_ϵ such that $|\Delta_E| \leq C_\epsilon N_E^{6+\epsilon}$ for elliptic curves $E/\mathbb{Q}$, where Δ_E is the minimal discriminant and N_E the conductor.*

The semistable case of the second conjecture already suffices.

Proof. Rational *abc* gives (1) by taking $1 + \epsilon = 121/100$. Oesterlé proves that the ordinary Szpiro discriminant–conductor inequality implies the weak *abc* inequality with exponent $6/5 + \epsilon$ for every $\epsilon > 0$ [1, Section I.3, p. 169]. Take $\epsilon = 1/100$. The stated implication uses the ordinary discriminant–conductor conjecture, not a separate inequality involving both elliptic invariants c_4 and c_6. □

2 Two consequences of the weak inequality

Put

$$\Theta = \left\{0, -\frac{1}{2}, -\frac{1}{4}, -\frac{2}{5}, -1, -\frac{1}{3}\right\}, \qquad S_0 = \{p : p \le 76176\}.$$

For $t \in \mathbb{Q} \setminus \Theta$, let $M_0(t)$ be the product of primes outside S_0 for which $v_p(t-\theta)$ is positive and odd for some $\theta \in \Theta$.

Lemma 2.1. *Assume (1). There is a constant $C_0 \ge 0$ such that*

$$h(t) \le \frac{121}{74} \log M_0(t) + C_0 \qquad (t \in \mathbb{Q} \setminus \Theta).$$

Proof. Consider the degree-four rational function

$$\varphi(z) = \frac{z(z+2)^3}{(2z+1)^3}, \qquad \varphi(z) - 1 = \frac{(z-1)^3(z+1)}{(2z+1)^3}.$$

Its complete inverse image of $\{0, 1, \infty\}$ is $\{0, -2, -1/2, 1, -1, \infty\}$. For a primitive integer pair (x, y) away from these six points, put

$$A = x(x+2y)^3, \quad C = y(2x+y)^3, \quad A - C = (x-y)^3(x+y),$$

$$G(x,y) = xy(x+2y)(2x+y)(x-y)(x+y).$$

The polynomial identity for $A - C$ follows by expansion. A prime dividing both A and C must divide both $x + 2y$ and $2x + y$, so it is 3. At that prime x, y are units and the smaller of those two valuations is one, because $2(x + 2y) - (2x + y) = 3y$. Hence $\gcd(A, C)$ is either 1 or 27. After division by this gcd, the three integers $A, C, A - C$ are pairwise coprime, and their radical is supported on G.

Write $H = \max(|x|, |y|)$. One has $\max(|x + 2y|, |2x + y|) \ge H$. If the corresponding prefactor $|x|$ or $|y|$ is at least $H/4$, this gives $\max(|A|, |C|) \ge H^4/4$. Otherwise the other prefactor is H and its linear factor is at least $H/2$, giving $\max(|A|, |C|) \ge H^4/8$. Consequently

$$h(\varphi(x/y)) \ge 4 \log H - \log 216.$$

The weak inequality therefore implies

$$\log \operatorname{rad}(G(x,y)) \ge \frac{4}{\kappa} \log H - \frac{\log 216 + \log K_\kappa}{\kappa}.$$

For a nonzero integer N, let $\operatorname{sf}(N)$ be the product of primes of odd valuation in N. Prime by prime, $\operatorname{rad}(N)^2 \le |N| \operatorname{sf}(N)$. Also $|G(x,y)| \le 36H^6$. Outside the fixed set S_0, two of its six primitive linear factors cannot vanish together. Thus its squarefree part outside S_0 is precisely the odd contact radical for the six displayed points. The contribution from S_0 is bounded by the fixed product of its primes. Combining these estimates gives

$$\log M(z) \ge \left(\frac{8}{\kappa} - 6\right) h(z) - C = \frac{74}{121} h(z) - C.$$

Finally $z = 2 + 1/t$ sends the six contacts in Θ to the six points above. Its integral matrix has determinant -1; its inverse is integral too, and both matrices have maximum row sum three. Hence it changes logarithmic height by at most $\log 3$. Substitution of a primitive numerator-denominator pair shows that the odd contact radicals agree outside S_0. This proves the lemma after adjusting the fixed constant. □

For a rational pair (u, v), define its projective height by

$$h_2(u, v) = h(1 : u : v) = \log \max(D_0, |A'|, |B'|),$$

where $D_0 > 0$, $u = A'/D_0$, $v = B'/D_0$ and $\gcd(D_0, A', B') = 1$. Both individual heights are at most h_2.

Lemma 2.2. *Assume (1). There is a constant $C_1 \geq 0$ such that*

$$h_2(u, v) \leq \frac{121}{437} h(u^8 + v^8) + C_1 \qquad (u, v \in \mathbb{Q}).$$

One may take $C_1 = \frac{279}{1748} \log 2 + \frac{100}{437} \log K_\kappa$.

Proof. If one coordinate is zero, $h_2(u, v) = h(u^8 + v^8)/8$, and if both are zero all heights vanish. Otherwise use the primitive representation above. Write $g = \gcd(|A'|, |B'|)$, $A_0 = A'/g$, $B_0 = B'/g$. Then $\gcd(A_0, B_0) = \gcd(g, D_0) = 1$. Put $H = \max(|A_0|, |B_0|)$ and $N = A_0^8 + B_0^8$; thus $H^8 \leq N \leq 2H^8$.

Let $k = \gcd(N, D_0^8)$, $m_0 = N/k$, and $n = D_0^8/k$. The fraction $u^8 + v^8 = m/n$ is reduced, where $m = g^8 m_0$. Put $M = \max(m, n)$, so the output height is $\log M$. Every prime dividing N either divides D_0 or survives in m_0, giving $\mathrm{rad}(N) \leq D_0 m_0$. Apply (1) to the pairwise coprime positive triple A_0^8, B_0^8, N:

$$H^{8/\kappa - 2} \leq K_\kappa^{1/\kappa} D_0 m_0.$$

Since $D_0^8 = nN/m_0 \leq 2nH^8/m_0$, it follows that

$$H^{437/121} \leq 2^{1/8} K_\kappa^{100/121} m_0^{7/8} n^{1/8}.$$

Moreover $g = (m/m_0)^{1/8}$ gives

$$\max(gH, D_0) \leq 2^{1/8} H (M/m_0)^{1/8}.$$

Taking logarithms and combining the last two inequalities yields

$$\begin{aligned} h_2(u, v) &\leq C_1 + \frac{205}{1748} \log m_0 + \frac{121}{3496} \log n + \frac{1}{8} \log M \\ &\leq C_1 + \frac{121}{437} \log M. \end{aligned}$$

Here $m_0, n \leq M$, all coefficients are nonnegative, and $205/1748 + 121/3496 + 1/8 = 121/437$. This accounts for the common numerator factor and every denominator cancellation. □

3 The complete rational source

Fix an odd integer $B \geq 3$ and set

$$\beta = 2X^2 + B. \tag{4}$$

For rational $X = u/v$ in lowest terms, integrality of β implies $v^2 \mid 2$, hence $v = 1$. Thus $\beta \in \mathbb{Z}$ if and only if $X \in \mathbb{Z}$. For integer input β is odd, and $\beta \geq B$ for every rational input.

Retain three pairs of rational coordinates, one for each of the points A, H, U. For any such pair (h_0, h_1) put

$$h = h_0 + h_1\sqrt{2}, \quad x = h^2, \quad j^2 = h^4 - D, \quad y = hj.$$

The direct point-chart theorem of [2] proves that these are exactly the nonzero multiples of P. Each finite choice j is encoded by one scalar monic quartic as follows. Write $h^4 - D = a + b\sqrt{2}$, put $c = 2a + b$, $d = -a - b$, and write $(c + d\sqrt{2})^2(a + b\sqrt{2}) = a' + b'\sqrt{2}$. Then

$$Y^4 - 4a'Y^2 + 8(b')^2 = 0, \quad j = \frac{Y/2 + (b'/Y)\sqrt{2}}{c + d\sqrt{2}}.$$

Its discriminant is $2^{15}(b')^2(2a^2 - b^2)^4(a^2 - 2b^2)^2$, nonzero at every rational pair (h_0, h_1). Indeed $h^4 - D \neq 0$ by its norm, the two quadratic norm factors are anisotropic over $\mathbb{Q}$, and $b' = -4a^3 + 6ab^2 + 3b^3$ is anisotropic by Eisenstein at 3. The square-root chart is complete because its transformed rational component is $2(u^3 - 2v^3)$ for $j = u + v\sqrt{2}$, which is nonzero for $j \neq 0$. These are algebraic facts independent of the height hypothesis.

Compute $V_0 = H - P$ and $V_1 = U - A$ by the usual affine addition formulas. Put $f = 1/x$, $z = -x/y$, let σ be conjugation of $K/\mathbb{Q}$, and define

$$L = \frac{z(U)}{z(H)}, \qquad L' = \sigma\left(\frac{z(V_1)}{z(V_0)}\right).$$

Use precisely the six field-valued tests

$$D_1 = f(H),\ D_2 = f(U),\ D_3 = \sigma f(V_0),\ D_4 = \sigma f(V_1),\ D_5 = L - L',\ D_6 = L - \beta. \tag{5}$$

Every test is assigned order one. No scalar-height pole or Lucas relation is present.

Let $T = \operatorname{Tr} x(A)$, and use the two positive rational scales

$$a_0 = 1, \qquad a_1 = \beta^8 + T^8.$$

Retain one rational slope λ. For $r = 0, 1$, set

$$t_r = a_r\lambda, \qquad \delta_r = t_r - 1, \qquad F_r = \prod_{\theta\in\Theta}(t_r - \theta), \qquad R_r = \frac{F_r}{\delta_r^8}, \tag{6}$$

$$W_r = \prod_{j=1}^{6}\left(1 + \frac{D_j^2}{\delta_r^4 R_r}\right), \qquad b_r = \frac{R_r}{\operatorname{Tr}(W_r^{-1})^2}. \tag{7}$$

All displayed divisions, affine-chart denominators, F_r, δ_r, $\operatorname{Norm}(W_r)$, and $\operatorname{Tr}(W_r^{-1})$ are guarded nonzero. These conditions are part of the source.

For each $b = b_r$, retain independent rational parameters τ_r, s_r. Put $B_0 = 17b$ and

$$e = \frac{\tau_r^2 - B_0}{\tau_r^2 + 3B_0}, \qquad q^2 = \frac{(s_r^2 - B_0e(B_0 - b^2))(s_r^2 + 3B_0e(B_0 - b^2))}{B_0}.$$

Guard the denominator and require $q \neq 0$. This reconstructs the first pole ordinate as $p = -4B_0\tau_r/(\tau_r^2 + 3B_0)$. Each rational-square equation $q^2 = N_0/D_0$ is made monic by $Z = D_0q$, giving $Z^2 = N_0D_0$ with $Z \neq 0$ and $D_0 \neq 0$. No new unrestricted rational parameter is used for these roots.

4 The local implication

Lemma 4.1. *For any rational solution of the guarded source, write $A = [n]P$, $H = [h]P$, $U = [u]P$. If $\beta \neq n$ and N is the positive absolute numerator of $\beta - n$ in lowest terms, then*

$$M_0(t_r) \mid N \qquad (r = 0, 1).$$

Proof. Fix a prime $p \mid M_0(t_r)$ and abbreviate $R = R_r$, $W = W_r$. Outside S_0 only one contact factor has positive valuation, δ_r is a unit, and $v_p(R)$ is odd because its denominator is an eighth power. For each place $\mathfrak{p}$ above p, the two terms of a factor of W have valuations of different parity. Hence

$$v_{\mathfrak{p}}(W) = \sum_j \min\{0, 2v_{\mathfrak{p}}(D_j) - v_p(R)\} \le 0.$$

Put $w = \max_{\mathfrak{p}|p} v_{\mathfrak{p}}(W) \le 0$. Then $v_p \operatorname{Tr}(W^{-1}) \ge -w$. The valuation of $b = R/\operatorname{Tr}(W^{-1})^2$ is odd, so the pole obstruction gives

$$0 < v_p(b) \le v_p(R) + 2w.$$

Thus $\nu = v_p(R) > 0$ and $w > -\nu/2$. At a place attaining w, each summand in the valuation of W is at least w. It follows that $v_{\mathfrak{p}}(D_j) > \nu/4 > 0$, so all six tests have valuation at least one at that place.

The first four tests put H, U into the formal group at $\mathfrak{p}$ and V_0, V_1 into it at the conjugate place. At these good unramified places the formal logarithm satisfies

$$\log_E(Q) = z(Q)(1 + O(p^4)).$$

Indeed its first omitted term is a fixed multiple of $Df(Q)^2$; $v_p(f(Q)) \ge 2$ in the formal group. The remaining coefficient estimates, uniformly for $p > 76176$, are those proved in the formal-logarithm paper [6]. The exact logarithmic ratios are u/h and $(u-n)/(h-1)$, respectively.

The affine guards imply $h \ne 0, 1$. At least one of $h, h-1$ is a p-adic unit, so the corresponding exact ratio is integral. Relative precision four makes its truncated ratio integral. The test $L - L' \equiv 0 \pmod p$ makes the other truncated ratio integral as well; the same relative precision then makes the other exact ratio integral. Relative errors are now absolute errors of valuation at least four. Consequently

$$L \equiv u/h, \qquad L' \equiv (u-n)/(h-1), \qquad L \equiv L' \pmod p.$$

Multiplying by the integral numbers $h, h-1$ and subtracting yields $L \equiv n \pmod p$. The target test gives $L \equiv \beta \pmod p$. In particular β is p-integral, and p divides the reduced numerator of $\beta - n$. This holds at every consumed prime and proves the divisibility. The same argument includes inert primes; if their formal-group requirements are incompatible, no source solution occurs at that prime. □

5 Soundness and the uniform input shift

Suppose a rational source solution has $\beta \ne n$. By Lemmas 4.1 and 2.1,

$$h(\lambda), h(a_1\lambda) \le \frac{121}{74} \log N + C_0.$$

The quotient-height inequality gives

$$h(a_1) \le \frac{121}{37} \log N + 2C_0.$$

Apply Lemma 2.2 to $\beta, T(n)$, and put

$$\eta = h_2(\beta, T(n)), \qquad q = \frac{14641}{16169} < 1, \qquad C = \frac{242}{437} C_0 + C_1.$$

Then

$$\eta \le q \log N + C. \tag{8}$$

All these constants depend only on the fixed contact map, the fixed power eight, and the weak-*abc* constant. They are independent of the shift B, the input, and every witness.

Writing $\beta = a/b$ in lowest terms gives $N = |a - nb| \le e^{h(\beta)}(1 + |n|)$, and $h(\beta) \le \eta$. Consequently

$$\eta \le c \log(1 + |n|) + C', \qquad c = \frac{q}{1-q}, \quad C' = \frac{C}{1-q}.$$

Since $h(T(n)) \le \eta$, the trace-height inequality implies

$$|n|^2 \le Ac \log(1 + |n|) + AC' + J \le Ac|n| + AC' + J.$$

Choose an integer $M > Ac + \sqrt{AC' + J}$. Then $|n| < M$, so

$$h(\beta) \le \eta \le c \log(1 + M) + C' =: H_*.$$

Choose $B_* > e^{H_*}$. For every fixed odd $B \ge B_*$, (4) gives $h(\beta) \ge \log \beta \ge \log B > H_*$, a contradiction. Thus $\beta = n \in \mathbb{Z}$, and the elementary input-map argument yields $X \in \mathbb{Z}$. This proves soundness uniformly for every sufficiently large shift. It does not determine that shift effectively.

6 Completeness

Fix an integer input X and an odd coefficient $B \ge 3$. Then $n = \beta$ is an odd integer at least three. Fix $A = [n]P$ and its two positive scales before choosing any prime. They are distinct because $a_1 > 1$. Temporarily put $H = [h]P$ and $U = [hn]P$.

Choose a nonexceptional $h_* \in 64\mathbb{Z}$, and work in the real orbit closure of $64\mathbb{Z}P$. Choose a compact real neighborhood for H around $[h_*]P$ on which the point coordinates, both difference charts, and the six tests are defined and bounded. Such a neighborhood exists: the rational functions restricted to the multiplication curve $H \mapsto (H, [n]H)$ are not identically singular. The difference charts exclude $h = \pm 1$, while every nonzero multiple of the non-torsion P has nonzero abscissa and ordinate. Pass to a smaller compact neighborhood to avoid all remaining zero denominators. Enlarge L with $64 \mid L$ and the real component-group condition so that the progression $h \equiv h_* \pmod{L}$ preserves fixed lower valuation bounds for these tests at $2, 3, 17$. These are precisely the local-continuity and real-orbit steps of the source realization, with fewer point roles and no scalar constraint.

Choose a fixed D_0 with $D_0^2\theta/a_r \in \mathbb{Z}$ for all contacts and both scales, set $v = D_0^2 w^2$, and seek $\lambda = u/v$. Apply the retained prime-pattern theorem to the distinct forms $u - (\theta/a_r)v$, after repetitions have been removed. All bad primes and previously fixed moduli are excluded before this application. The theorem permits arbitrarily large divisibility of w, a sufficiently large prescribed lower bound on u/w^2, and the conjugate reduction-order conditions.

As in [3, 4], the annihilating order moduli U_0, V_0 for the two conjugate families satisfy

$$\gcd(U_0, V_0) = 1, \qquad \gcd(U_0, L) = 16, \qquad \gcd(V_0, L) = 1.$$

Thus the congruences $h \equiv 0 \pmod{U_0}$, $h \equiv 1 \pmod{V_0}$, $h \equiv h_* \pmod{L}$ are compatible, since $16 \mid h_*$. Choose h in that progression while retaining the real neighborhood and avoiding the finitely many excluded multiples. At one place above each contact prime, H, U are formal; at the conjugate place, $H - P, U - A$ are formal. Both exact logarithmic ratios are n. The constant approximation has precision four, so all six tests have their assigned order one, including the target $n = \beta$.

The inherited detector preparation now applies without alteration. For clarity, its main ingredients are the following. The identity

$$R_r = \frac{a_r^6 \prod_{\theta\in\Theta}(u-(\theta/a_r)v)}{v^6(a_r\lambda-1)^8}$$

has a rational-square prefactor. In particular, the possibly nonsquare scale contributes the square a_r^6. At each chosen contact prime $v_p(R_r)=1$; the successful conjugate place gives all inverse norm-factor residues equal to one, and the trace of W_r^{-1} is a unit with residue one or two. The resulting b_r has square reduced denominator, and the prescribed congruences give its local squareclasses at $2,3,17$. The fixed lower bounds on the tests and sufficient divisibility of w give the depths in (2). On the compact real set, uniformly,

$$R_r \sim a_r^{-2}\lambda^{-2}, \qquad W_r = 1+O(\lambda^{-2}), \qquad b_r \sim (4a_r^2\lambda^2)^{-1}.$$

Thus $0<b_r<17$ for the sufficiently large lower bound on λ fixed before prime selection. All detector guards are nonzero. These calculations are the source comparison-pole preparation proof [4]; they depend on the finite list of tests and their orders, not on a Lucas or scalar test.

Each prepared input has an independent pole certificate with $pqs\neq 0$ by the retained pole-realization hypotheses. Since $p\neq 0$ excludes $e=1$, the substitution $\tau=p/(e-1)$ gives the single chart used above and its nonzero denominator. Thus both nonzero quadratic-root conditions are realized. The direct point charts cover A,H,U. Every rational input $X\in\mathbb{Z}$ therefore has all required witnesses, for every fixed odd $B\geq 3$.

7 The polynomial, its bounds, and verification

The retained coordinates are exactly

Coordinates	Number
Input X	1
Three point pairs for A,H,U	6
Comparison slope λ	1
Two independent pole pairs (τ_r,s_r)	4
Retained before finite-root elimination	12

There are three monic point quartics and two monic pole quadratics. The point algebra has rank $4^3=64$, and the full algebra has rank $4^3 2^2=256$. Its complete guard is gq_1q_2 with g in the point algebra, and each quadratic radicand also belongs to that algebra. There are no further residual equations. The separating-element theorem [7] therefore replaces all five roots and all nonzero guards by one rational coordinate, without changing the input projection. This gives thirteen total coordinates. Its determinant formulas and its 64-factor product, applied to the supplied integer coefficient circuit, define F_B completely.

The native generator weights are

$$6,6,6,51131,69765.$$

The full guard and its point factor have weights 148317 and 27421. In the notation of [7, Theorem 4.1], the multiplication-element weight is at most $W_1 r+W_0$, where

$$W_1 = 9577692, \qquad W_0 = 70636907.$$

Thus

$$\deg F_B \leq 256(9577692\cdot 64^2+(70636907+1)\cdot 127) = 12339485119488 < 2^{44}.$$

The input shift is a named coefficient parameter of weight zero in the coefficient graph. No operation or branch depends on its numerical value. Consequently these weight inequalities, and hence the degree bound, hold symbolically for every fixed B.

The directory `computation/few_variables/weak_abc13` contains the rational-source builder, its integer coefficient specification, and independent source and bound certificates. The source has 1758 rational-circuit nodes and its integer coefficient graph has 754 nodes. The independent conversion auditor checks every affine substitution, the three point quartics and their automatic nonzero factors, both pole-square clearings, and all 48 atomic guard factors by exact expression identities. The separate finite-algebra checker verifies coefficient dependencies, companion-column inequalities and all 64 degree cases without importing the builder.

For the algebraic fixture $B = 3$ it certifies

$$\log_2 H(F_3) \le 29454018907833 < 2^{45}.$$

This is not a height bound for a member certified to define $\mathbb{Z}$. The proof provides an ineffective sufficient shift, whereas the coefficient algorithms construct and bound each specified member. The exact finite checks establish their recorded algebraic assertions; they do not verify (1) or the inherited arithmetic premises.

7.1 A uniform bound on the number of monomials

An ordinary monomial involves the polynomial coordinates after all fixed coefficient parameters have been substituted and equal monomials have been collected. The determinant construction is not expanded here, so its exact number of nonzero monomials is not known.

Proposition 7.1. *Suppose the construction of [7] uses n total coordinates, a point algebra of rank b, a full algebra of rank d, and a total degree bound D. Put $E = (2b-1)d$. If $E < D$, its number of nonzero ordinary monomials is at most*

$$\binom{D+n}{n} - \binom{D-E+n-1}{n}.$$

For $E \ge D$ the bound is simply $\binom{D+n}{n}$.

Proof. Each determinant $H_r(U) = \det(I - UM_{\beta_r})$ has degree at most d in the additional coordinate U, on every specialization. The final polynomial is one factor H_b times $b-1$ factors $H_r^2 + \sum_{j>r} a_j^2$. Thus its degree in U is at most E; the remaining summands have no U dependence. There are $\binom{D+n}{n}$ monomials of total degree at most D in n coordinates. Those with U exponent at least $E+1$ correspond, by subtracting $E+1$ from that exponent, to monomials of degree at most $D-E-1$. Subtracting their number proves the formula. Actual coefficient cancellations can only decrease the support. □

For the present family, $b = 64$, $d = 256$ and $E = 32512$. The exact integer bound is less than 2^{509}, uniformly in the fixed shift. Applying the same argument to the 27- and 29-variable specifications of [7] gives strict upper bounds 2^{2202} and 2^{2417}, respectively. The independent script `computation/few_variables/check_monomial_bounds.py` checks the three binomial-coefficient calculations with exact integers. These are upper bounds; none is asserted to be an exact support count.

8 Other weak exponents

Corollary 8.1. *Retain the three arithmetic inputs of Section 1. Suppose that the weak rational abc inequality holds for one fixed exponent $1 < \kappa < 4/3$. Choose an even integer $d \ge 4$ such*

that

$$\left(\frac{8}{\kappa}-6\right)\left(\frac{d}{\kappa}-3\right)>2.$$

Replacing the second scale by $a_1=\beta^d+T^d$ gives a computable family of polynomials in thirteen total variables whose sufficiently large fixed odd shifts define $\mathbb{Z}$ in $\mathbb{Q}$. Such a choice of d exists for every stated κ.

Proof. The contact calculation in Lemma 2.1 gives

$$h(t)\le\gamma^{-1}\log M_0(t)+O(1),\qquad \gamma=8/\kappa-6>0.$$

Put $\alpha=d/\kappa-3$. Repeating the primitive numerator–denominator calculation in Lemma 2.2, with d in place of eight, gives

$$h_2(u,v)\le\alpha^{-1}h(u^d+v^d)+O(1).$$

Indeed, the three coefficients of $\log m_0,\log n,\log M$ in that calculation become $(d-1-\alpha)/(d\alpha)$, $1/(d\alpha)$ and $1/d$. They are nonnegative and sum to $1/\alpha$. The two local radical divisibilities therefore imply

$$h_2(\beta,T(n))\le\frac{2}{\gamma\alpha}\log N+O(1).$$

The coefficient is less than one by the displayed choice of d. The proof of soundness consequently gives a bound independent of the fixed shift, and a sufficiently large odd shift excludes $\beta\neq n$.

Evenness of d makes a_1 positive and greater than one. The six contacts, the square prefactor a_1^6, all local tests, and the completeness argument are unchanged. There are still twelve retained coordinates, three monic point quartics and two monic pole quadratics. The same finite-algebra theorem therefore produces a thirteenth coordinate and an exact finite coefficient algorithm for each chosen d. □

The numerical degree and coefficient bounds in the preceding section are certified for the supplied specialization $\kappa=121/100$, $d=8$. No numerical degree bound for other values of d is asserted here.

References

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